SQE Lab 2: ISO 25010 Quality Attribute Analysis

Section 1: Cover Information

- **App Names:** WhatsApp vs. Messenger
- **Version Numbers:** WhatsApp (2.26.14.74) | Messenger (554.0.0.59.70)
- **Platform:** Android
- **OS Version:** 1.0.16.0.UCVMIXM
- **Evaluator Names:** Tayyab Farooq, Nabeel Abid
- **Student IDs:** FA23-BSE-064, FA23_BSE-052
- **Pair Letter:** A
- **Date of Evaluation:** 2026-04-22
- **Device Model:** Redmi 12C

Section 2: App Overviews

WhatsApp: Developed by WhatsApp LLC, this app is a globally dominant instant messaging and Voice-over-IP (VoIP) application. Its primary target audience includes general consumers and professionals seeking a reliable, low-bandwidth communication tool. Its main features include end-to-end encrypted text messaging, voice notes, video calls, and file sharing. It falls strictly under the Communication category, prioritizing streamlined utility.
Facebook Messenger: Developed by Meta Platforms, Inc., Messenger is a feature-rich messaging platform deeply integrated with the broader Facebook social ecosystem. Its primary purpose is to connect users with their social network, offering rich media sharing, stories, custom chat themes, and seamless cross-device synchronization.
While both apps share the same fundamental purpose of instant messaging, WhatsApp focuses heavily on minimalistic, hardware-efficient utility, whereas Messenger prioritizes dynamic social features and high-level personalization.

Section 3: Evaluation Methodology

The evaluation was conducted using a standardized 1–5 Likert rating scale to assess all eight ISO 25010 software quality characteristics. Testing was performed collaboratively using a single Redmi 12C device connected to a standard Wi-Fi network. This ensured both applications were subjected to the exact same hardware constraints and network conditions. The evaluation took approximately 50 minutes per app. A digital stopwatch was used to accurately measure performance efficiency operations (such as cold boot times and file sending speeds) in actual seconds, rather than relying on subjective estimates.

Section 4: Completed Checklists

App 1: WhatsApp (Evaluator: Tayyab Farooq & Nabeel Abid)

Characteristic	Weight	Rating (1-5)	Justification (Evidence)
Functional Suitability	15%	5	**Obs 1:** Text messages are sent and received reliably without delay. **Obs 2:** Voice messages are recorded, sent, and heard properly with clear audio quality.
Performance Efficiency	12%	4	**Obs 1:** Cold boot took exactly 2.1 seconds on the Redmi 12C. **Obs 2:** Sending a 5MB image took 1.8 seconds over Wi-Fi.
Compatibility	10%	4	**Obs 1:** Co-exists well with other background apps without forcing closures. **Obs 2:** Call audio switches seamlessly when transitioning between networks or handling standard cellular calls.
Usability	15%	5	**Obs 1:** The interface is highly intuitive; core settings are easily discoverable. **Obs 2:** Navigation is straightforward with no cluttered menus, minimizing cognitive load for new users.
Reliability	15%	5	**Obs 1:** The application experienced zero crashes or freezes during the testing period. **Obs 2:** Messages remained perfectly synced and intact even after toggling Airplane mode.
Security	15%	4	**Obs 1:** End-to-end encryption is active by default for all chats. **Obs 2:** However, chat backups to Google Drive require manual encryption setup, posing a slight risk for unobservant users.

Characteristic	Weight	Rating (1-5)	Justification (Evidence)
Maintainability	8%	4	**Obs 1:** The app receives frequent, seamless background updates via the Play Store. **Obs 2:** Bug fixes are pushed regularly with clear, standardized release notes.
Portability	10%	3	**Obs 1:** Account migration requires a localized backup or Google Drive. **Obs 2:** Moving chat history across different OS platforms (e.g., Android to iOS) is historically complex and requires third-party tools or tedious localized transfers.

App 2: Facebook Messenger (Evaluator: Tayyab Farooq & Nabeel Abid)

Characteristic	Weight	Rating (1-5)	Justification (Evidence)
Functional Suitability	15%	4	**Obs 1:** Core chat features work well. **Obs 2:** However, the heavy integration with Facebook "Stories" clutters the core messaging utility, distracting from the primary purpose.
Performance Efficiency	12%	3	**Obs 1:** Cold boot took 3.4 seconds on the Redmi 12C. **Obs 2:** Navigating between heavy media chats and GIF libraries caused noticeable stuttering and frame drops on this specific hardware.
Compatibility	10%	4	**Obs 1:** Excellent "Chat Heads/Bubbles" integration with the Android OS. **Obs 2:** This native API hook allows for highly effective multitasking over other running applications.
Usability	15%	3	**Obs 1:** The main interface is heavily saturated with secondary tabs. **Obs 2:** Navigating to

Characteristic	Weight	Rating (1-5)	Justification (Evidence)
			specific privacy settings or core account features requires navigating through confusing, nested menus.
Reliability	15%	4	**Obs 1:** The app is generally stable with no hard crashes during testing. **Obs 2:** Minor interface hanging occurred when switching rapidly between heavy chat threads.
Security	15%	3	**Obs 1:** End-to-end encryption is not enabled by default for all standard chats. **Obs 2:** Users must manually navigate menus and explicitly initiate "Secret Conversations" to secure their data from server-side storage.
Maintainability	8%	4	**Obs 1:** Regular updates are consistently available via the Play Store. **Obs 2:** Internal error logging and crash reporting are handled seamlessly without user intervention.
Portability	10%	5	**Obs 1:** Excellent portability due to purely cloud-based architecture. **Obs 2:** Users can log in on any device instantly without relying on manual backups or localized device hubs.

Section 5: Quality Profile Chart

(When submitting your PDF, ensure you copy and paste the image file ***image_64d1bf.png*** *that you generated via Python directly into this section. It serves as your official visual proof of the comparative analysis.)*

Section 6: Comparative Analysis

Upon completing a rigorous ISO 25010 evaluation on the Redmi 12C hardware, WhatsApp achieved a superior weighted quality score of 4.35 out of 5.00, compared to Facebook Messenger’s score of 3.68. This results in a decisive margin of 0.67 points in

favor of WhatsApp. This quantifiable difference highlights fundamentally different development philosophies between the two Meta-owned properties. WhatsApp is engineered as a lightweight, high-reliability utility that scales perfectly down to budget hardware. In contrast, Messenger is designed as a feature-dense social hub, which inherently penalizes its overall quality score when constrained by the processing limitations of an entry-level smartphone.

The most critical quality gaps between the two applications were observed in Usability, Portability, and Performance Efficiency. The Usability gap was the most significant, with WhatsApp scoring a perfect 5 and Messenger scoring a 3. For users in practice, this means Messenger's interface—cluttered with "Stories," "Communities," and heavy media tabs—increases cognitive load and makes finding core messaging features frustrating. Conversely, Messenger dominated the Portability gap (scoring 5 vs. WhatsApp's 3). Because Messenger is entirely cloud-based, a user can instantly log into a secondary device and access their entire message history without relying on localized device backups. WhatsApp's heavy reliance on the primary phone as a local hub severely limits a user's freedom to seamlessly jump between different operating systems. Finally, the Performance gap heavily favored WhatsApp. On the Redmi 12C, Messenger’s cold boot took 3.4 seconds and suffered from UI stuttering, whereas WhatsApp booted in a snappy 2.1 seconds. For a user checking their phone dozens of times a day, those extra seconds of latency and stuttering in Messenger compound into a noticeably degraded daily experience.

A distinct and vital trade-off is evident in this evaluation: Functional Suitability (Feature Richness) versus Performance Efficiency. Facebook Messenger actively attempts to maximize its Functional Suitability by offering a vast array of secondary functions, including custom chat themes, interactive games, and deep social media integration. However, on a budget device like the Redmi 12C, this richness acts as a double-edged sword; it directly degrades Performance Efficiency by demanding excess RAM and CPU cycles, causing the observed slower load times and UI stuttering. WhatsApp makes the conscious trade-off to strip away these complex, resource-heavy features in order to maintain a highly responsive, fluid, and reliable experience across all hardware tiers.

Based on this comprehensive quality profile, I highly recommend WhatsApp for users operating in low-bandwidth geographical areas or those utilizing entry-level hardware like the Redmi 12C. Its unparalleled reliability, speed, and default security make it the optimal choice for critical daily communication. Conversely, Facebook Messenger is recommended specifically for users who prioritize deep social features, rich media communication, and multi-device synchronization, provided they possess a mid-tier or flagship device with sufficient RAM to handle the app's heavy resource overhead without system lag.

Section 7: Recommendations

1. WhatsApp: Improving Portability through Cloud Architecture
WhatsApp should implement seamless, purely cloud-based multi-device syncing without requiring the primary phone to be online or act as a localized hub in order to improve

Portability from its current rating of 3 to a target of 5. Currently, account migration and cross-platform syncing heavily rely on localized backups or complex Google Drive transfers, which historically break down when moving from Android to iOS. This structural change would immensely benefit users by giving them true device independence, allowing them to instantly access their secure chat history on any tablet, desktop, or new phone without fear of data loss or technical hurdles.

2. WhatsApp: Enhancing Security for Cloud Backups

WhatsApp should make end-to-end encrypted chat backups the mandatory default setting during initial account setup to improve **Security** from a 4 to a perfect 5. Currently, while live chats are encrypted natively, users must manually navigate into their settings to turn on encryption for their Google Drive backups. This leaves a significant loophole where non-technical users might unknowingly have their chat histories stored in plain text on cloud servers. Enforcing this by default would ensure that user data remains absolutely secure both in transit and at rest, eliminating the weakest link in their privacy architecture.

3. Messenger: Mandating Default Encryption

Facebook Messenger should enable end-to-end encryption by default across all standard chat threads to improve **Security** from its current unacceptable rating of 3 to a target of 5. Currently, standard chats are stored server-side, and users are forced to manually open separate "Secret Conversations" to secure their data from potential third-party interception. This change would dramatically benefit users by ensuring all personal communication is intrinsically safe by default, granting them the exact same baseline level of privacy and peace of mind that WhatsApp currently provides, without requiring them to possess any technical knowledge.

4. Messenger: Streamlining the User Interface

Facebook Messenger should introduce a dedicated 'Lite' view toggle within the settings that completely disables the "Stories" and "Community" tabs to improve **Usability** from a 3 to a target of 5. Currently, the main interface is visually cluttered and saturated with non-essential social features, which causes navigation confusion and contributes directly to slower load times on entry-level devices like the Redmi 12C. This interface change would provide a massive benefit to the user base by offering a streamlined, distraction-free environment for individuals who strictly want to utilize the app for its core, fundamental messaging functions.

GitHub Repository Link:https://github.com/tayyabfarooq005/sqe-labs-tayyab-farooq.git